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Roll No

EC-8004 (1) (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Advanced Digital Signal Processing

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
 ii) All questions carry equal marks.

1. a) Define Z-transform and derive the Z transform of
 $f(nT) = \sin \omega nT$. 7
 b) Determine DTFT of $x[n] = (a^n \sin \Omega_0 n) u[n]$. 3
 c) Determine DTFT of $x[n] = \left(\frac{1}{4}\right)^n u[n]$. 4
2. a) Describe the basic principle of spectrum estimation in detail. 7
 b) Find out the response of linear system to random signals. 7
3. Explain and derive the formula of DFT and write its properties. 14
4. a) Explain radix-4 FFT algorithm in detail. 7
 b) Explain the linear convolution using DFT. 7
5. a) Explain the design of IIR and FIR digital filters. 7
 b) Give the comparison between FIR and IIR filters. 7
6. a) Explain the relationship between filter parameters and auto correlation sequence. 7
 b) Explain the implementation for sampling rate conversion. 7
7. Determine the z-transform of the signal
 a) $x(n) = (\cos \omega_0 n) u(n)$ 7
 b) $x(n) = (\sin \omega_0 n) u(n)$ 7
8. a) Differentiate between DTFT and DFT. 6
 b) Explain Least Square Method. 4
 c) Explain Power Spectrum. 4
